

Probabilities on Graphs: Spatial Context

Markov random fields and mean field theory
Notes for Probabilistic Models of the Visual Cortex

Alan Yuille

Johns Hopkins University

17 September, 2026

What this lecture covers

From pixels to spatial context

Markov random fields

Long-range interactions

The inference problem

Mean field theory

Summary

From directed graphs to undirected graphs

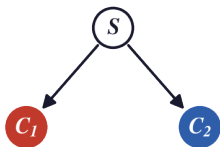
In a directed graph each node carries a conditional distribution given the parents of the node: the two-cue model $S \rightarrow C_1, S \rightarrow C_2$ has $P(S)$ at the root and $P(C_1 | S), P(C_2 | S)$ at the leaves, and the joint is their product.

In an undirected graph each node carries evidence and each edge a potential that scores compatibility. No node is the cause of another; the edge (i, j) says only that the two labels x_i and x_j constrain each other. The joint distribution is built from the potentials on the nodes and the edges, and it has to be normalised over every configuration at once.

The new question is: when every pixel is a variable, how do neighbouring labels constrain each other, and how is the labelling inferred?

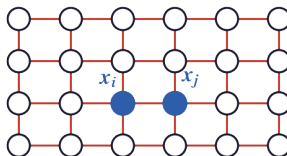
From directed graphs to undirected graphs

directed graph



each edge points
from cause to cue

undirected graph



the labels on an edge
must be compatible

Figure 1: The two-cue model of the earlier lecture and the pixel grid of this one. An arrow points from cause to cue; an undirected edge joins two labels that must be compatible.

Semantic segmentation labels every pixel

The image I is observed. The unknown is the labelling

$$x_i \in \{\text{airplane, sky, person, ...}\}, \quad i \in W, \quad \mathbf{x} = \{x_i : i \in W\},$$

one label for every pixel i in the set of pixels W . In the segmentation deck the image is a pair of cyclists, and the target labelling paints every pixel as person, bicycle or background.

The number of pixels is large, so $\mathbf{x}$ has tens of thousands of components to estimate together. The model of this lecture treats them as one random vector rather than as $|W|$ separate decisions.

Semantic segmentation labels every pixel

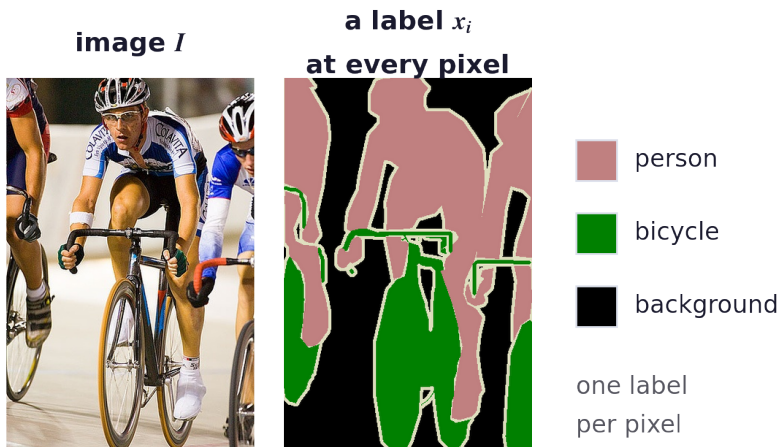


Figure 2: An image and its labelling, one label per pixel (Chen et al., 2015).

A deep network scores every label at every pixel

A convolutional neural network (CNN) gives a score for every label at every output position. Applied fully convolutionally (Long et al., 2015), it turns an image into one score map per label, so the airplane score map is high on the fuselage and low on the sky.

Each output position sees one region of the image, the receptive field of that position. The score maps are the evidence for the labels; the spatial model takes the score maps as its input.

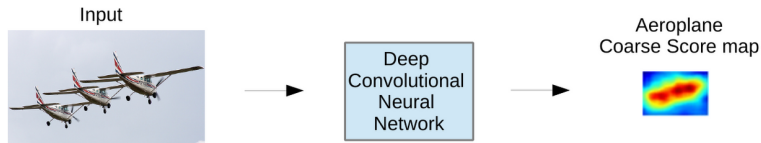


Figure 3: The first stage of the DeepLab pipeline: the image enters a deep convolutional network and a coarse score map for each label comes out (Chen et al., 2015).

Recognition needs context, localisation needs detail

A large receptive field lets the network recognise the horse from its whole shape and surroundings. The same receptive field blurs the boundary: the score map misses the outline of the rider and the legs, because every output position averages over a window much larger than the boundary.

The score map is therefore right in the interior of an object and coarse at its boundary. Chen et al. (2015) call this the accuracy and localisation tradeoff, and it is the reason a spatial model is added after the network.

Recognition needs context, localisation needs detail

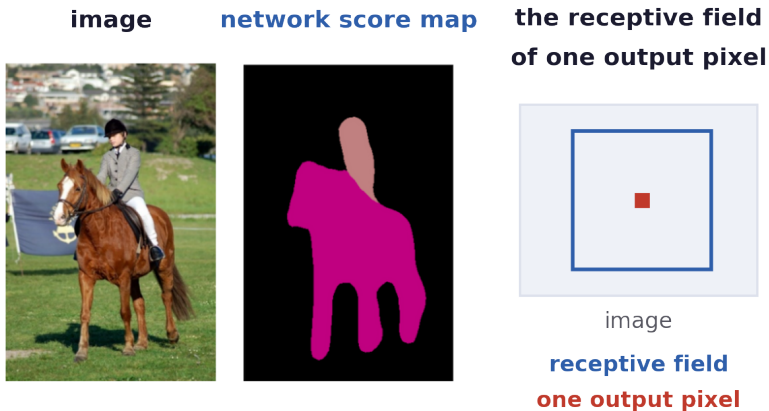


Figure 4: The score map of the network for the horse, and the receptive field of one output pixel. The window that makes recognition possible is what blurs the boundary (images: Chen et al., 2015).

Neighbouring labels constrain each other

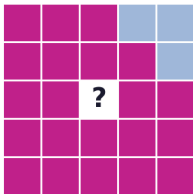
A pixel can be ambiguous from the image alone; the labels of the neighbouring pixels are further evidence. Deciding each pixel on its own uses the product of the per-pixel posteriors, and deciding them jointly uses the joint distribution of the whole labelling:

$$\prod_{i \in W} P(x_i | I) \quad \text{against} \quad P(x | I),$$

the pixels on their own against the joint labelling. The labels are inferred jointly, from the image at each pixel and from the labels of the neighbours. The move from $P(x_i | I)$ to $P(x | I)$ is the whole content of the lecture: the second distribution has to say how the labels depend on one another.

Neighbouring labels constrain each other

pixel by pixel



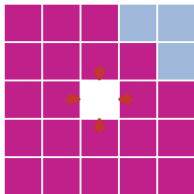
the centre is ambiguous
on its own

x_i depends on
the image

and on the labels
 x_j next to it



jointly with its neighbours



the neighbours favour
the horse label

Figure 5: The centre pixel is ambiguous on its own. Read together with its four neighbours, which all say horse, it is a horse pixel.

One node for every pixel, one edge for every neighbouring pair

A Markov random field (MRF) is an undirected graph with one node for the label of each pixel in W . The neighbours of a pixel are the pixels next to it,

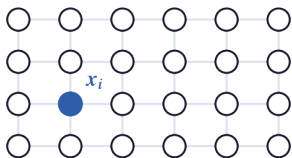
$$N(i) = \{\text{the pixels next to } i\}, \quad \text{one edge } (i,j) \text{ for every } j \in N(i),$$

and on the image grid $N(i)$ is the set of four pixels above, below, to the left and to the right of i .

If one pixel is an airplane then the neighbouring pixels are likely to be airplanes too. An edge means the two labels interact: the pairwise term scores how compatible the two labels are. The word Markov refers to the property this graph gives the distribution: conditioned on the labels of its neighbours, the label of a pixel is independent of every other label.

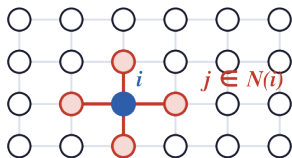
One node for every pixel, one edge for every neighbouring pair

one node for every pixel



W is the set of pixels,
 x_i the label of pixel i

edges join neighbouring pixels



$N(i)$: the four
neighbours of pixel i

Figure 6: The nodes of the field are the pixels; the edges join each pixel to the four pixels in $N(i)$.

The energy scores a labelling: evidence plus context

The energy $E(x)$ gives every labelling x of the image one number; a low energy is a good labelling.

$$E(x) = \underbrace{\sum_{i \in W} \psi_i(x_i)}_{\text{unary evidence}} + \underbrace{\sum_{(i,j) \in \mathcal{N}} \psi_{ij}(x_i, x_j)}_{\text{context}},$$

where $\mathcal{N} = \{\text{neighbouring pairs } (i,j)\}$ is the set of edges. Two kinds of terms: a unary term $\psi_i(x_i)$ for each pixel, which depends on the label of that pixel alone, and a pairwise term $\psi_{ij}(x_i, x_j)$ for each neighbouring pair, which depends on the two labels at the ends of the edge. Each pair is counted once, so the second sum runs over the set of edges $\mathcal{N}$ rather than over i and then over $j \in N(i)$, which would count every edge twice.

Both terms are costs. The unary term says what the image says at pixel i ; the pairwise term says whether the labels at the two ends of the edge are compatible.

The energy scores a labelling: evidence plus context

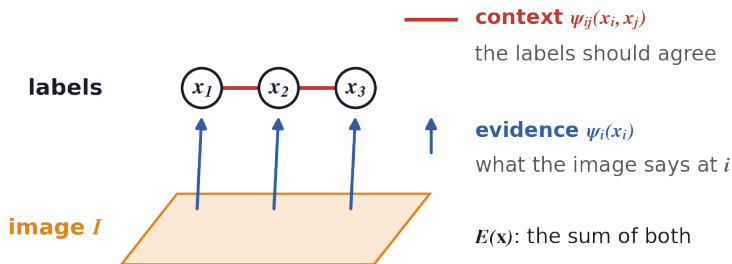


Figure 7: The structure of the energy. Blue arrows carry the evidence from the image to each label; red edges carry the context between neighbouring labels.

The unary term is the evidence from the deep network

The unary term is a cost: it is low where the network gives the label a high score and high elsewhere,

$$\psi_i(x_i) = -\log P(x_i | I),$$

where $P(x_i | I)$ is the network's posterior for the label x_i at pixel i , read off the score map after a softmax over the labels. The minus log turns a probability into a cost, so a high probability is a low energy: a pixel in the middle of the fuselage has $P(\text{airplane} | I)$ near one and a unary cost near zero, and a pixel on the sky pays a large cost for the airplane label.

The unary evidence can be learned by a deep network; the spatial model takes it as given. Nothing in the spatial model depends on how the scores were produced, only on the fact that they are available at every pixel for every label.

The unary term is the evidence from the deep network

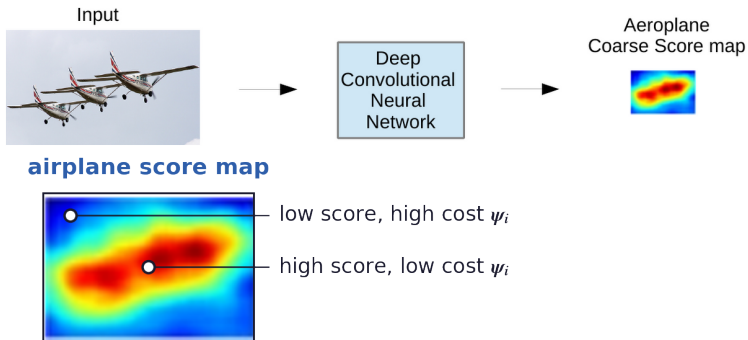


Figure 8: The airplane score map of the network, enlarged. A pixel on the fuselage has a high score and a low unary cost; a pixel on the sky has a low score and a high unary cost (score map: Chen et al., 2015).

The pairwise term charges a cost when neighbours disagree

The Potts model is the simplest choice: agreement is free and every disagreement costs the edge weight,

$$\psi_{ij}(x_i, x_j) = w_{ij} \mu(x_i, x_j),$$

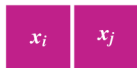
$$\mu(x_i, x_j) = 1 \text{ if } x_i \neq x_j, \quad \mu(x_i, x_j) = 0 \text{ if } x_i = x_j.$$

The function μ is the label compatibility, written in this form by Krähenbühl and Koltun (2011). The edge weight $w_{ij} > 0$ can be a constant for the whole image or can depend on the two pixels.

With a constant weight w , the context term of the energy is w times the number of neighbouring pairs whose labels differ, which is w times the length of the boundaries between the regions of the labelling. Lowering the energy therefore shortens the boundaries.

The pairwise term charges a cost when neighbours disagree

same label



$\psi_{ij} = 0$

different labels



$\psi_{ij} = w_{ij}$

a disagreement
between neighbours

costs w_{ij}

Figure 9: The Potts term on one edge: zero when the two labels agree, w_{ij} when they differ.

The best labelling has the lowest energy

The labelling estimated from the model is the one with the lowest energy,

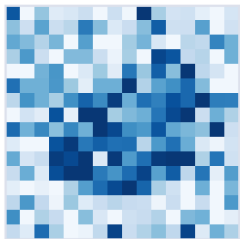
$$\hat{x} = \arg \min_x E(x).$$

On a 16 by 16 binary grid with the unary cost equal to minus the log of a noisy score map and a Potts cost with $w_{ij} = 1$: labelling A takes at every pixel the label with the higher score, so it follows the evidence pixel by pixel and pays for every isolated disagreement; its unary part is 73 and it has 178 disagreeing neighbour pairs, giving $E(A) = 251$. Labelling B, reached from A by repeatedly setting each pixel to the label that lowers the energy given its neighbours, pays a little more in evidence, 111, and far less in context, 49 disagreeing pairs, so $E(B) = 160$.

Labelling B is the better segmentation and the energy says so, which is the reason to score whole labellings rather than pixels.

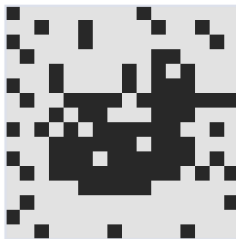
The best labelling has the lowest energy

score map $P(x_i=I|I)$



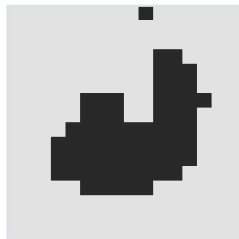
the evidence

labelling A



$E(A) = 251$

labelling B



$E(B) = 160$

Figure 10: The score map, the labelling that follows it pixel by pixel, and a labelling with lower energy.

A conditional random field conditions on the image

An MRF is a distribution over the labels; a conditional random field (CRF) conditions on the image. In segmentation the image I is observed, so both terms of the energy may depend on the image,

$$E(\mathbf{x}; I) = \sum_{i \in W} \psi_i(\mathbf{x}_i; I) + \sum_{(i,j) \in \mathcal{N}} \psi_{ij}(\mathbf{x}_i, \mathbf{x}_j; I).$$

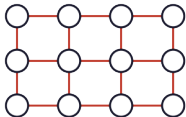
The unary term already depends on the image through the network. The pairwise term may depend on it too, for example through a weight w_{ij} that is large when the two pixels have the same colour.

A CRF models $P(\mathbf{x} | I)$ directly and never needs a distribution over images. Because the image is fixed throughout inference, the notes write $E(\mathbf{x})$ and $P(\mathbf{x})$ from here on, with the dependence on I understood.

A conditional random field conditions on the image

Markov random field

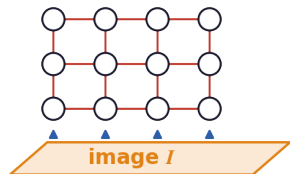
$$P(\mathbf{x})$$



a distribution over
the labels alone

conditional random field

$$P(\mathbf{x}|I)$$



the terms of the energy
depend on the image

Figure 11: A Markov random field over the labels alone, and a conditional random field whose terms depend on the image.

A grid model joins each pixel to its four neighbours

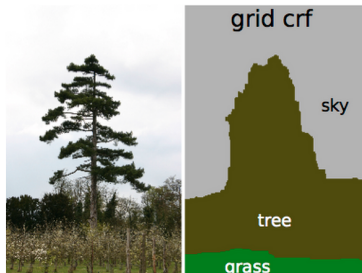
The grid model is the energy with $\mathcal{N}$ the set of four-neighbour pairs,

$$E(\mathbf{x}) = \sum_{i \in W} \psi_i(x_i) + \sum_{(i,j) \in \mathcal{N}} \psi_{ij}(x_i, x_j), \quad \mathcal{N} : \text{ the four-neighbour pairs.}$$

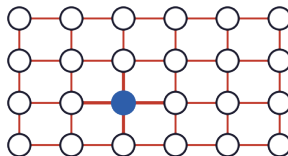
Approximate inference on the grid is fast: about one second for fifty thousand variables, the timing reported by Krähenbühl and Koltun (2011). The interactions are local, so the model has limited expressive power and smooths the labels. Distant pixels never interact directly; they influence each other only through the chain of neighbours between them.

A grid model joins each pixel to its four neighbours

image and the grid-model labelling



each pixel talks to its four neighbours



distant pixels never
interact directly

Figure 12: The grid model labelling of the tree image, and the four edges of one pixel (labelling: I. Kokkinos).

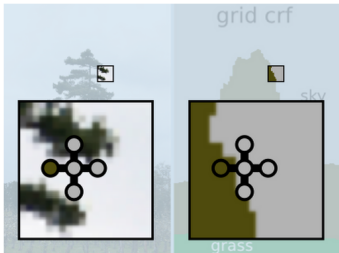
Constant weights smooth across the object boundary

The four edges at a boundary pixel look exactly like the four edges inside the object. With a constant weight every edge pulls the two labels together, so thin branches are smoothed away: a branch one pixel wide has two disagreeing edges for every pixel of its length, and removing it saves that context cost at the price of a few unary costs.

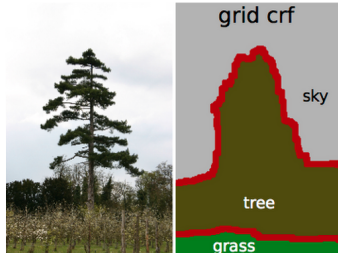
This is the shrinking bias of the grid model. The tree keeps its trunk and loses its branches, and the boundary with the grass is pulled into a straight line.

Constant weights smooth across the object boundary

**the edges
at a boundary**



**grid model:
shrunk boundary**



a constant w_{ij} pulls every pair of labels
together, across the boundary too

thin branches are lost: the shrinking bias

Figure 13: The edges at a boundary and the grid model result: the thin branches are lost (figures: I. Kokkinos).

The image sets the strength of each edge

Nearby pixels of similar colour are coupled strongly; a change of colour weakens the coupling,

$$\psi_{ij}(x_i, x_j) = w_{ij}(I) \mu(x_i, x_j).$$

The edge weight is read from the image, so a colour boundary becomes a weak edge in the model. A disagreement between a tree pixel and a sky pixel then costs little, and the labelling is free to follow the outline of the tree.

The image sets the strength of each edge

similar colour



w_{ij} **large**

different colour



w_{ij} **small**

the pair is judged
by the image itself:

w_{ij} **depends on I**

Figure 14: The same edge on two pairs of pixels: a strong coupling where the colours match and a weak one where they differ.

A fully connected model joins every pair of pixels

Every node is connected to every other node, and the connections are weighted differently,

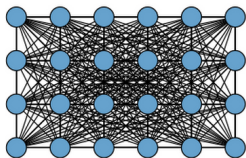
$$E(\mathbf{x}) = \sum_{i \in W} \psi_i(\mathbf{x}_i) + \sum_{i \in W} \sum_{j > i} \psi_{ij}(\mathbf{x}_i, \mathbf{x}_j),$$

where the condition $j > i$ counts each pair once. The interaction is strongest between pixels that are close in position and similar in appearance, and it falls off smoothly with distance and colour difference. This is the fully connected CRF of Krähenbühl and Koltun (2011).

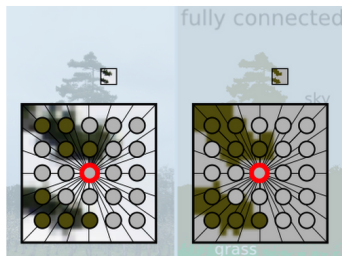
The number of edges is $|W|(|W| - 1)/2$, about 10^9 for an image of fifty thousand pixels, so the model can only be used if inference never visits the edges one by one.

A fully connected model joins every pair of pixels

all pairs are joined



the edges of one pixel



dark edges: strong coupling to pixels
that look like this one

light edges: weak coupling to the rest

Figure 15: A fully connected graph and the edges of one pixel on the tree image: dark edges couple it to pixels that look like it (figures: I. Kokkinos).

Nearby pixels of similar colour interact most strongly

The edge weight is a sum of two Gaussian kernels on the positions p_i and the colours l_i of the two pixels,

$$w_{ij} = w_1 \exp\left(-\frac{\|p_i - p_j\|^2}{2\sigma_\alpha^2} - \frac{\|l_i - l_j\|^2}{2\sigma_\beta^2}\right) + w_2 \exp\left(-\frac{\|p_i - p_j\|^2}{2\sigma_\gamma^2}\right).$$

The first kernel compares the positions and colours of the two pixels: close and similar, strong coupling. The second kernel is a weak smoothing term over position alone. The scales σ_α , σ_β and σ_γ set how far and over what colour difference the coupling reaches.

The kernel maps of the figure were computed on the tree image with $\sigma_\alpha = 50$ pixels and $\sigma_\beta = 0.3$ in colour units scaled to $[0, 1]$, with $w_1 = 1$ and $w_2 = 0$: a pixel on the tree is coupled to the tree and a pixel on the grass to the grass.

Nearby pixels of similar colour interact most strongly



bright: nearby and the same colour
the coupling follows the outline of the tree,
so the labels keep that outline

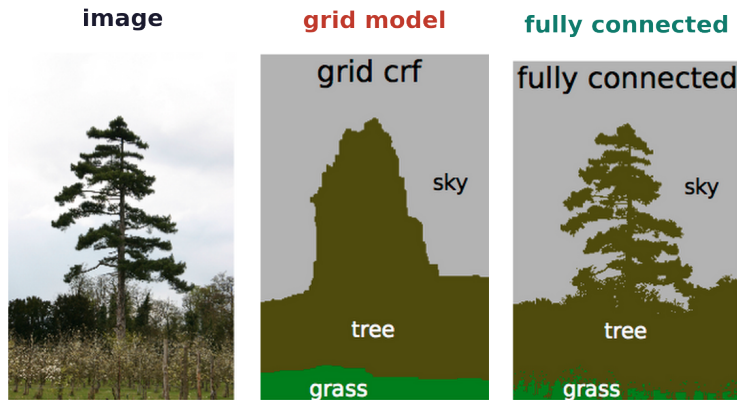
Figure 16: The first kernel for a pixel on the tree and a pixel on the grass. Bright means nearby and the same colour.

Weighted long-range interactions recover the branches

The grid model kept the trunk and lost the branches; the fully connected model recovers both. A pixel on a branch is coupled to every other pixel of the tree with the same colour, however far away, so the branch label is supported by the whole tree and no longer by two neighbours only.

The grass line is placed where the colour changes, since that is where the edges are weak. The shrinking bias is gone because the context term no longer measures the length of the boundary.

Weighted long-range interactions recover the branches



thin branches and the grass line are recovered

long-range interactions, no shrinking bias

Figure 17: The image, the grid model labelling and the fully connected labelling (figures: I. Kokkinos, from Krähenbühl and Koltun, 2011).

The CRF aligns the network's labels with the image

The two pieces are put together. The coarse score map is interpolated to the image size and becomes the unary term,

$$E(x) = \sum_i \psi_i(x_i) + \sum_i \sum_{j>i} \psi_{ij}(x_i, x_j), \quad \psi_i(x_i) = -\log P_{\text{network}}(x_i | I),$$

and the pairwise term is the Gaussian edge potential of the fully connected model. The fully connected CRF aligns label boundaries with colour boundaries in the image; this is DeepLab (Chen et al., 2015).

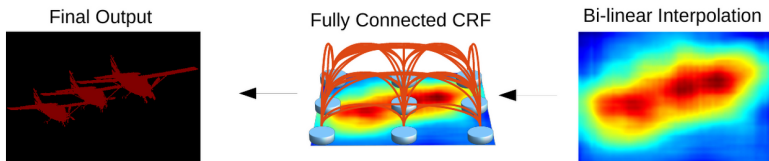
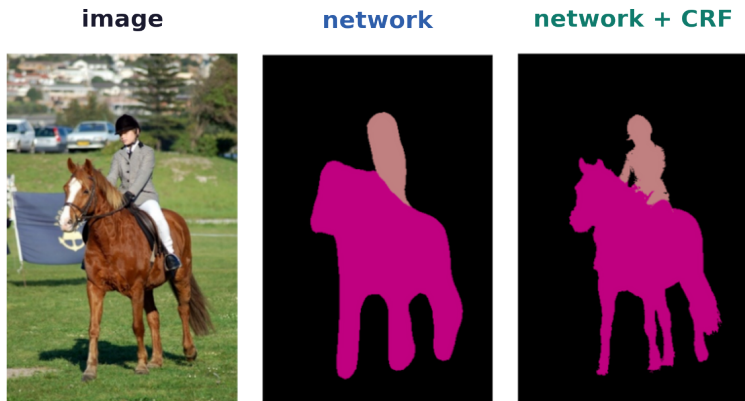


Figure 18: The second stage of the DeepLab pipeline: the score map is interpolated to the image size and the fully connected CRF produces the final labelling (Chen et al., 2015).

The CRF recovers the boundary of the horse

The score map of the network is spatially coarse around the rider and the legs. The CRF, using the colours of the image, follows the boundaries of rider, horse and legs more closely: the unary term still decides that the region is a horse, and the pairwise term decides where the horse ends.

The CRF recovers the boundary of the horse



the score map is coarse around rider and legs

the CRF follows the boundaries in the image

Figure 19: The image, the labelling from the network alone, and the labelling after the CRF (Chen et al., 2015).

The CRF refines the object boundaries

In each example the network finds the object and the CRF refines the boundary of the object. Thin structures, the bicycle and the wings, are where the difference is largest, because a thin structure is exactly what a large receptive field blurs.

On the PASCAL VOC 2012 benchmark the gain is measured by the mean intersection over union (IoU) between the predicted and the true regions, averaged over the classes. Earlier methods reached about 50 percent; the deep network alone reaches 60 to 64 percent; the deep network with the fully connected CRF reaches above 67 percent; and pretraining the network on the larger COCO dataset takes the combination to 71 percent (Chen et al., 2015).

The CRF refines the object boundaries

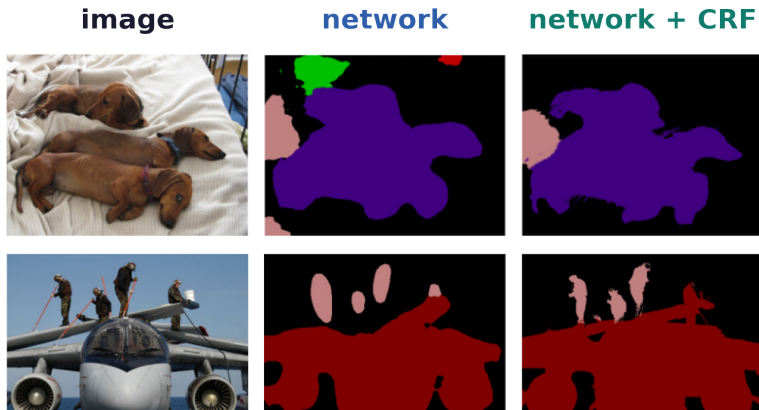


Figure 20: A dog and an airplane: image, network labelling, labelling after the CRF (Chen et al., 2015).

The CRF refines the object boundaries

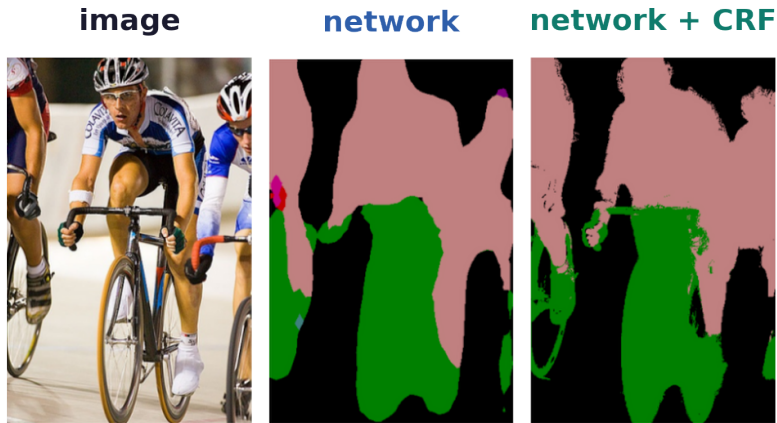


Figure 21: Cyclists: the thin frame of the bicycle is where the CRF changes the labelling most (Chen et al., 2015).

The number of labellings grows exponentially

If each pixel takes M states, there are M to the power $|W|$ configurations of x :

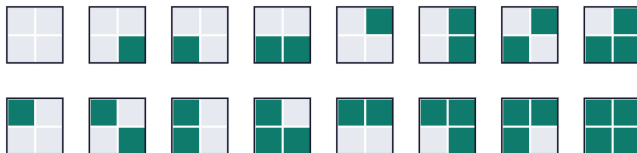
$|W|$ pixels, M labels each : $M^{|W|}$ configurations of x .

Four pixels with two labels give $2^4 = 16$ labellings, which can be drawn on a page. An image of fifty thousand pixels with twenty-one labels gives 21^{50000} , far too many to enumerate.

Finding the best labelling is typically very hard: NP-hard for a general graph. A tree still allows exact inference by dynamic programming, in time linear in the number of nodes; on a grid with loops exact inference is intractable in general, and the algorithms of this lecture are approximations.

The number of labellings grows exponentially

four pixels, two labels: $2^4=16$ labellings



every labelling of the four pixels

an image with $|W|$ pixels and M labels:

$M^{|W|}$ labellings

too many to enumerate

Figure 22: The sixteen labellings of four binary pixels, and the count for an image.

The energy defines a probability distribution

The Gibbs distribution gives every labelling a probability; low energy is high probability,

$$P(x) = \frac{e^{-E(x)}}{Z}, \quad Z = \sum_x e^{-E(x)}.$$

The normalising constant Z sums over all labellings, so for a large model Z is typically out of reach. The image is fixed throughout, so from here on $P(x)$ is short for $P(x|I)$ and $E(x)$ for $E(x; I)$.

For the four-pixel example, with the score map $(0.8, 0.6, 0.3, 0.2)$ read as $P(x_i=1|I)$, unary costs $-\log P$, and a Potts cost $w_{ij} = 0.7$ on the four edges of the 2×2 grid, the sixteen energies run from 2.71 to 6.74, the sum $Z = \sum_x e^{-E(x)}$ is 0.288, and the probabilities run from 0.23 down to 0.004. The Gibbs form sends a difference of one unit of energy to a factor $e \approx 2.7$ in probability.

The energy defines a probability distribution

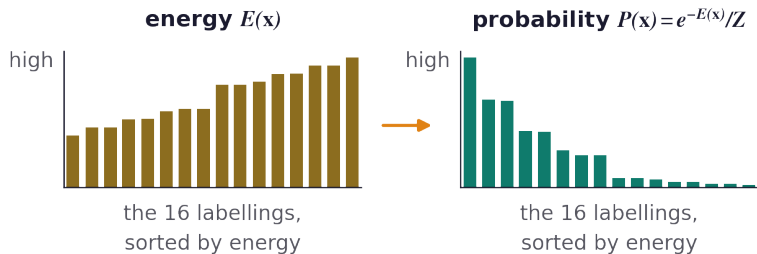


Figure 23: The sixteen energies of the four-pixel example and the probabilities they define, in the same order.

Minimising the energy is maximising the probability

Because e^{-E} decreases with E and Z is the same for every labelling, the lowest-energy labelling is the most probable labelling,

$$\hat{x} = \arg \min_x E(x) = \arg \max_x P(x),$$

the MAP (maximum a posteriori) estimate. Segmentation can be treated as energy minimisation or probabilistic inference; the optimum is the same. In the four-pixel example the best labelling puts the top two pixels, whose scores are 0.8 and 0.6, in the object and the bottom two outside it: its unary part is 1.31, it has two disagreeing edges costing 1.4, and $E = 2.71$, $P = 0.023$.

The probabilistic form adds something the energy alone lacks: the marginal $P(x_i)$ of one pixel, which is a sum of $P(x)$ over every other pixel, says how confident the model is at i , and mean field theory estimates exactly these marginals.

Minimising the energy is maximising the probability

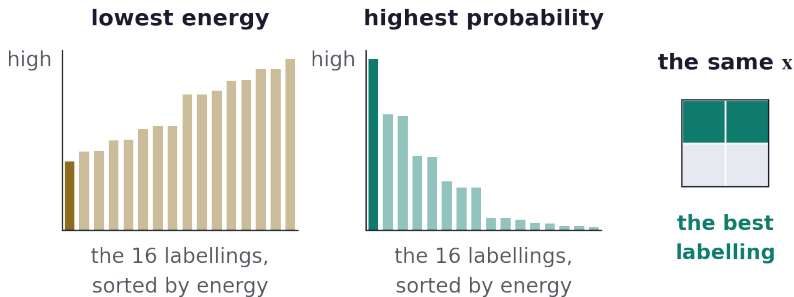


Figure 24: The lowest bar of the energy and the highest bar of the probability are the same labelling.

Three approaches to inference on the Gibbs distribution

Sampling by Markov chain Monte Carlo (MCMC), mean field theory (MFT), and belief propagation (BP).

- ▶ MCMC draws samples $x^{(1)}, x^{(2)}, \dots$ from $P(x)$ by a random walk over labellings, and estimates marginals and the best labelling from the samples.
- ▶ MFT replaces $P(x)$ by a distribution that factorises over the pixels and fits it by minimising a free energy.
- ▶ BP passes messages along the edges of the graph; on a tree it is exact, on a loopy graph it is an approximation.

This course describes MFT and BP; a handout (Yuille chapter) describes all three. MCMC is almost always much slower than MFT and BP.

Three approaches to inference on the Gibbs distribution

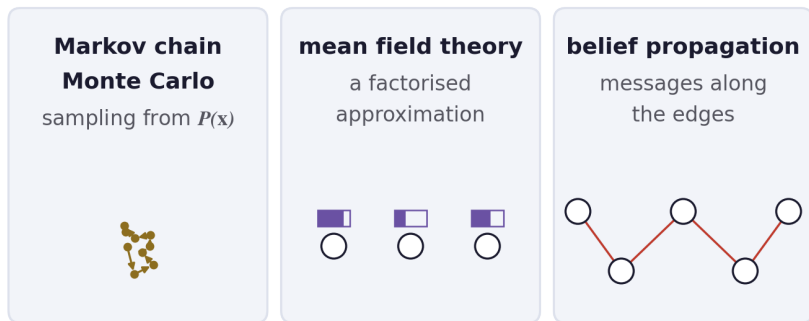


Figure 25: Three approximate inference algorithms: a random walk over labellings, one belief per pixel, messages along the edges.

The true distribution couples every pixel to its neighbours

Take three binary pixels in a chain, $x_i \in \{0, 1\}$, with

$$P(x_1, x_2, x_3) = \frac{1}{Z} \exp[-\psi_1 x_1 - \psi_2 x_2 - \psi_3 x_3 - \psi_{12} x_1 x_2 - \psi_{23} x_2 x_3],$$

$$\psi_1 = -1.2, \quad \psi_2 = 0.4, \quad \psi_3 = -0.8, \quad \psi_{12} = \psi_{23} = -1.$$

A negative ψ_i favours $x_i = 1$ and a negative ψ_{ij} rewards the two pixels for being on together. Three binary pixels in a chain already need eight probability entries, one for each joint state; in the order 000, 001, ..., 111 they are 0.016, 0.036, 0.011, 0.066, 0.054, 0.121, 0.099, 0.597.

The most probable state is a joint configuration of all three pixels, 111, and the middle pixel is on in it although its own evidence $\psi_2 = 0.4$ favours off: the context from both neighbours outweighs the evidence. The marginals $P(x_i=1)$ are 0.870, 0.773 and 0.820.

The true distribution couples every pixel to its neighbours



three binary pixels

$2^3 = 8$ **joint states**

one number for each

the joint distribution $P(x_1, x_2, x_3)$

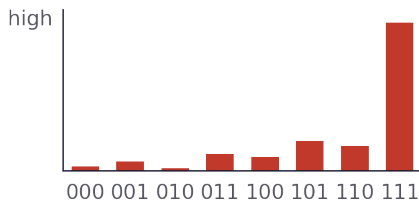


Figure 26: Three coupled pixels and the eight probabilities of their joint states.

Mean field approximates P by independent factors

Mean field theory replaces $P(x)$ by a distribution that factorises,

$$Q(x) = \prod_{i \in W} q_i(x_i), \quad \hat{x}_i = \arg \max_{x_i} q_i(x_i), \quad \hat{x} = \{\hat{x}_i : i \in W\}.$$

Each factor q_i is a distribution over the labels of one pixel, the belief at pixel i . The approximation Q assumes independence; the true P is coupled and stays coupled. Q is factorisable, so the maximum of Q is found pixel by pixel: the label with the largest belief.

For the three-pixel chain, mean field theory gives beliefs $q_i(x_i=1)$ of 0.879, 0.787 and 0.830 against the true marginals 0.870, 0.773 and 0.820. The labelling read off the beliefs is 111, the same as the most probable joint state. The beliefs are slightly too confident, which is typical of the approximation: Q cannot represent the correlation between the pixels, so it puts the probability of the coupled state into the individual marginals.

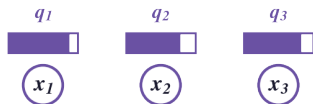
Mean field approximates P by independent factors

$P(\mathbf{x})$: the pixels are coupled



eight joint probabilities

$Q(\mathbf{x}) = q_1(x_1) q_2(x_2) q_3(x_3)$



three numbers, one per pixel,
no edges

Figure 27: The coupled distribution needs eight numbers; the factorised approximation keeps three beliefs, one per pixel, and no edges.

Choose the factorised distribution closest to P

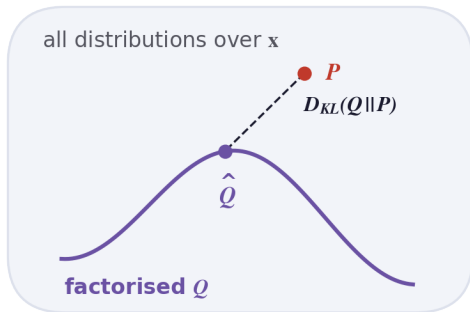
The criterion is the Kullback-Leibler (KL) divergence of Q from P ,

$$\hat{Q} = \arg \min_Q \sum_x Q(x) \log \frac{Q(x)}{P(x)},$$

minimised over the factorised distributions. The KL divergence is zero when $Q = P$ and positive otherwise, so it measures how far the approximation is from the truth. The order of Q and P matters: the expectation is taken under Q , which is the distribution that can be summed over, because it factorises; the expectation under P would need the sum over all labellings that the approximation is meant to avoid.

The search is restricted to factorised distributions; within that family the closest one is chosen. The true P lies outside the family, so the minimum is positive, and the minimiser $\hat{Q}$ is the best independent account of a distribution that is not independent.

Choose the factorised distribution closest to P



the true P is coupled
and lies off the curve

$\hat{Q}$: the point of the
curve closest to P
in $D_{KL}(Q \parallel P)$

Figure 28: Among the factorised distributions, the one nearest to P in the KL divergence.

The KL divergence splits into a free energy and log Z

Writing the KL divergence as two sums,

$$F[Q] = \sum_x Q(x) \log Q(x) - \sum_x Q(x) \log P(x),$$

and substituting $\log P(x) = -E(x) - \log Z$ from the Gibbs distribution, the second sum becomes $\sum_x Q(x) E(x) + \log Z \sum_x Q(x)$, and $\sum_x Q(x) = 1$, so

$$F[Q] = \underbrace{\sum_x Q(x) \log Q(x)}_{\text{negative entropy}} + \underbrace{\sum_x Q(x) E(x)}_{\text{expected energy}} + \underbrace{\log Z}_{\text{the same for every } Q}.$$

The $\log Z$ term is independent of Q , so $\log Z$ is never needed when $F[Q]$ is minimised: the sum over all labellings that defines Z has dropped out of the problem. The two Q -dependent terms, expected energy minus entropy, form the variational free energy, and minimising the KL divergence over Q is the same as minimising the variational free energy.

Since the KL divergence is never negative, the variational free energy is never below $-\log Z$, and it equals $-\log Z$ only when $Q = P$.

A continuous approximation replaces the discrete search

The minimisation has changed character,

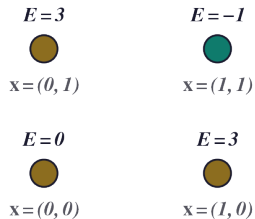
$$\hat{x} = \arg \min_x E(x) \text{ (discrete)} \quad \longrightarrow \quad \hat{Q} = \arg \min_Q F[Q] \text{ (continuous)}.$$

$F[Q]$ is a differentiable function of continuous variables, the beliefs, so $F[Q]$ can be minimised downhill. For two binary pixels with $\psi_1 = \psi_2 = 3$ and $\psi_{12} = -7$, the energy takes the values 0, 3, 3 and -1 on the four labellings (0,0), (1,0), (0,1) and (1,1), while $F[Q]$ is a smooth surface over the unit square of beliefs (q_1, q_2).

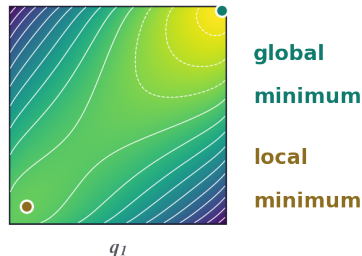
$F[Q]$ is differentiable but in general has several local minima; the search is restricted to factorised Q . On this surface the global minimum is at (0.98, 0.98) with $F = -1.04$ (leaving out $\log Z$), near the corner of the best labelling (1,1), and there is a second, local minimum at (0.08, 0.08) with $F = -0.12$, near the corner (0,0): each pixel alone prefers to be off, and only together do they prefer to be on. A descent started near (0,0) stays there.

A continuous approximation replaces the discrete search

$E(x)$ on four labellings



$F[Q]$ on the square



a search over discrete points

Figure 29: The energy on the four labellings of two binary pixels, and the free energy on the continuous square of beliefs, with its global and its local minimum.

Binary example: the free energy in terms of the beliefs

Take binary labels and the energy of the chain example on a general graph,

$$x_i \in \{0, 1\}, \quad E(x) = \sum_i x_i \psi_i + \sum_{(i,j) \in \mathcal{N}} x_i x_j \psi_{ij},$$

$$q_i(x_i=1) = q_i, \quad q_i(x_i=0) = 1 - q_i.$$

For this binary example the energy is bilinear in x ; the expectation under Q replaces labels by beliefs. Under a factorised Q the pixels are independent, so $\sum_x Q(x) x_i = q_i$ and $\sum_x Q(x) x_i x_j = q_i q_j$, and the expected energy is $\sum_i q_i \psi_i + \sum_{(i,j) \in \mathcal{N}} q_i q_j \psi_{ij}$. The entropy of a factorised distribution is a sum over pixels, because $\log Q$ is a sum of the $\log q_i$. Together,

$$F[Q] = \sum_i \{ q_i \log q_i + (1 - q_i) \log(1 - q_i) \} \\ + \sum_i q_i \psi_i + \sum_{(i,j) \in \mathcal{N}} q_i q_j \psi_{ij} + \log Z.$$

The evidence is weighted by one belief, the context by the product of two.

The entropy of a factorised distribution is a sum over pixels

For $Q(x) = \prod_i q_i(x_i)$ the logarithm is a sum, $\log Q(x) = \sum_i \log q_i(x_i)$, so

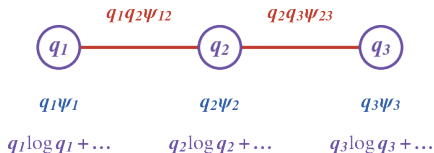
$$\sum_x Q(x) \log Q(x) = \sum_i \sum_x Q(x) \log q_i(x_i) = \sum_i \sum_{x_i} q_i(x_i) \log q_i(x_i),$$

because summing $Q(x)$ over every pixel other than i leaves the marginal $q_i(x_i)$. For binary labels the inner sum has two terms,

$$\sum_{x_i} q_i(x_i) \log q_i(x_i) = q_i \log q_i + (1 - q_i) \log(1 - q_i),$$

which is the negative entropy of one pixel: it is $-\log 2 \approx -0.69$ at $q_i = 1/2$ and rises to 0 at $q_i = 0$ or 1. The negative entropy of Q is therefore the sum of $|W|$ such terms, one per pixel, and it favours uncertain beliefs while the expected energy favours confident ones.

Binary example: the free energy in terms of the beliefs



entropy of each pixel

evidence, weighted by q_i

context, weighted by $q_i q_j$

$\log Z$: the same for every Q

Figure 30: The terms of the binary free energy on a chain of three pixels: an entropy at each node, the evidence weighted by one belief, the context weighted by the product of the two beliefs on the edge.

Each pixel updates its belief from its neighbours' beliefs

With the other beliefs fixed, the derivative in one belief vanishes at a sigmoid of evidence and beliefs. Differentiating $F[Q]$ with respect to q_i , with every q_j for $j \neq i$ held fixed,

$$\frac{\partial F[Q]}{\partial q_i} = \log q_i - \log(1 - q_i) + \psi_i + \sum_{j \in N(i)} q_j \psi_{ij} = 0,$$

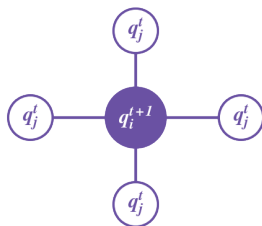
where the pairwise sum collects every edge that has i at one end. Solving for q_i gives $\frac{q_i}{1-q_i} = e^{-\psi_i - \sum_j q_j \psi_{ij}}$ and hence

$$q_i^{t+1} = \frac{e^{-\psi_i - \sum_j q_j^t \psi_{ij}}}{1 + e^{-\psi_i - \sum_j q_j^t \psi_{ij}}} = \sigma\left(-\psi_i - \sum_{j \in N(i)} q_j^t \psi_{ij}\right),$$

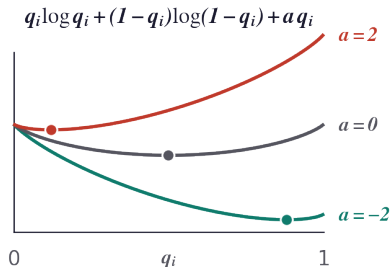
with $\sigma(u) = 1/(1 + e^{-u})$ the sigmoid. Initialise from the unary evidence, $q_i^0 = \sigma(-\psi_i)$, and repeat the update: the iteration seeks a stationary point of $F[Q]$. Writing $a = \psi_i + \sum_j q_j^t \psi_{ij}$ for the total cost of switching pixel i on, the per-pixel part of $F[Q]$ is

$q_i \log q_i + (1 - q_i) \log(1 - q_i) + a q_i$, and its minimum over q_i is at $\sigma(-a)$: at $a = -2$ it is 0.88, at $a = 0$ it is 1/2, at $a = 2$ it is 0.12.

Each pixel updates its belief from its neighbours' beliefs



beliefs in,
new belief out



each minimum is at $q_i = \sigma(-a)$

Figure 31: The beliefs of the neighbours enter the update of one pixel, and the per-pixel part of the free energy for three values of the total cost a , each minimised at $\sigma(-a)$.

The update generalises to the fully connected model

With M labels the belief of a pixel becomes a vector over the labels, and the sigmoid becomes a softmax. For the fully connected model with the Gaussian edge potentials, Krähenbühl and Koltun (2011) write the update as

$$Q_i(x_i=l) = \frac{1}{Z_i} \exp \left\{ -\psi_i(l) - \sum_{l'} \mu(l, l') \sum_m w_m \sum_{j \neq i} k^m(f_i, f_j) Q_j(l') \right\},$$

where $k^m(f_i, f_j)$ is the m -th Gaussian kernel evaluated on the features $f_i = (p_i, l_i)$ of the two pixels, and Z_i normalises the belief over the labels l .

The sum over all pixels j is a Gaussian filtering of the current beliefs, so a dense graph is still fast.

The update generalises to the fully connected model

For a fixed kernel m and label l' , computing $\sum_{j \neq i} k^m(f_i, f_j) Q_j(l')$ for every pixel i is one Gaussian filtering of the belief map $Q_j(l')$, so one iteration of the update runs in four steps.

1. Message passing: for each kernel m and label l' , compute $\tilde{Q}_i^m(l') = \sum_{j \neq i} k^m(f_i, f_j) Q_j(l')$, a Gaussian filter of the belief map $Q_j(l')$ in the feature space.
2. Compatibility transform: $\hat{Q}_i(l) = \sum_{l'} \mu(l, l') \sum_m w_m \tilde{Q}_i^m(l')$.
3. Local update: $Q_i(l) \propto \exp\{-\psi_i(l) - \hat{Q}_i(l)\}$.
4. Normalise each Q_i over the labels.

The update generalises to the fully connected model

On the MSRC-21 dataset (591 images, 21 classes), the same unary evidence under three models gives, in global pixel accuracy: the unary term alone 84.0; the grid CRF 84.6, in one second per image; the fully connected CRF 86.0, in 0.2 seconds per image, for about fifty thousand pixels. The average per-class accuracy rises from 76.6 to 77.2 to 78.3. Mean field on the dense model is faster than the grid model because the Gaussian filtering is done by a fast approximate algorithm whose cost is linear in the number of pixels, and MCMC on the same fully connected model took 36 hours (Krähenbühl and Koltun, 2011).

The update generalises to the fully connected model

MSRC dataset

- 591 images
- 21 classes

	Time	Global	Avg
Unary	-	84.0	76.6
Grid CRF	1s	84.6	77.2
FC CRF	0.2s	86.0	78.3

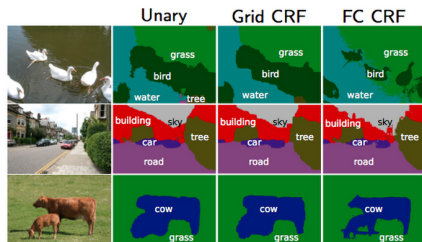


Figure 32: The three models on the same unary evidence: accuracy, time, and example labellings (Krähenbühl and Koltun, 2011).

Mean field iterations sharpen the segmentation

Iteration 0 is the evidence of the network; each iteration refines the beliefs with weighted messages from the pixels that are close and similar in colour. After ten iterations the belief map follows the outline of the airplane, and the labelling read off the beliefs matches the ground truth far more closely than the labelling read off the score map.

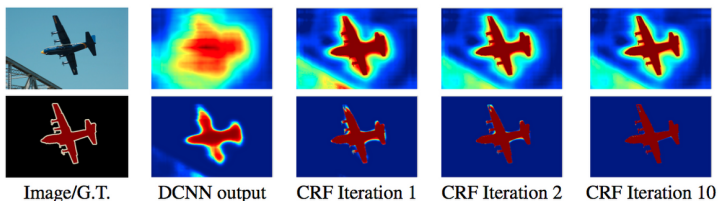


Figure 33: The score map of the network, then the belief map and its labelling after one, two and ten mean field iterations (Chen et al., 2015).

The concave-convex procedure: each update lowers F

Split the binary free energy into a convex and a concave part,

$$F[q] = E_{\text{convex}}[q] + E_{\text{concave}}[q],$$

$$E_{\text{convex}}[q] = \sum_i \{ q_i \log q_i + (1 - q_i) \log(1 - q_i) \},$$

$$E_{\text{concave}}[q] = \sum_i q_i \psi_i + \sum_{(i,j) \in \mathcal{N}} q_i q_j \psi_{ij}.$$

The entropy part is convex, because

$\frac{d^2}{dq^2} [q \log q + (1 - q) \log(1 - q)] = \frac{1}{q(1-q)} > 0$. The second part is linear plus bilinear, and it is concave when the matrix of pairwise terms is negative semidefinite; a quadratic term can enforce concavity, since adding $\lambda \sum_i q_i^2$ to the convex part and subtracting it from the second part, with λ large enough, makes the second part concave without changing F (Yuille and Rangarajan, 2003).

The concave-convex procedure: each update lowers F

Linearising the concave part at the current beliefs gives an upper bound on F , because a concave function lies below its tangent. Minimising the bound sets the gradient of the convex part at the new point equal to minus the gradient of the concave part at the old one,

$$\left. \frac{\partial E_{\text{convex}}}{\partial q_i} \right|_{q^{t+1}} = - \left. \frac{\partial E_{\text{concave}}}{\partial q_i} \right|_{q^t}, \quad \log \frac{q_i^{t+1}}{1 - q_i^{t+1}} = -\psi_i - \sum_{j \in N(i)} q_j^t \psi_{ij},$$

where the second form substitutes the two derivatives, and it is the sigmoid update. So each step lowers F : the concave-convex procedure (CCCP) explains why the mean field iteration converges. Under suitable conditions the updates reach a local minimum of F ; they cannot find the global minimum in general, since F can have several.

In one variable, with $E_{\text{convex}} = q \log q + (1 - q) \log(1 - q)$ and $E_{\text{concave}} = -q - 3q^2$: at $q^t = 0.25$ the slope of the concave part is $-1 - 6 \times 0.25 = -2.5$, so the update sets $\log \frac{q}{1-q} = 2.5$ and $q^{t+1} = \sigma(2.5) = 0.92$, and F falls from -1.00 to -3.76 in that one step.

The concave-convex procedure: each update lowers F

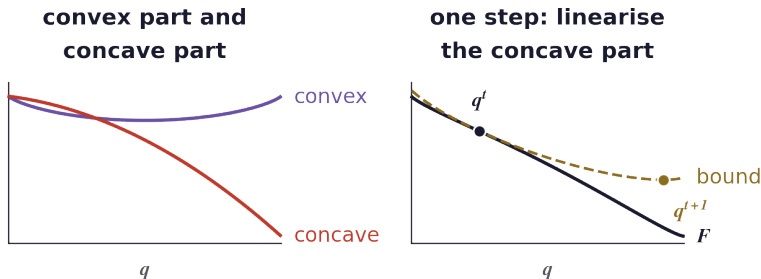


Figure 34: The convex and concave parts of a one-variable free energy, and one step of the procedure: the tangent to the concave part gives a bound whose minimum is the next belief.

Deterministic annealing starts from a smoother distribution

Introduce a temperature T into the Gibbs distribution and into the free energy,

$$P(x; T) = \frac{e^{-E(x)/T}}{Z_T},$$

$$F[Q, T] = \sum_x Q(x) \log Q(x) + \frac{1}{T} \sum_x Q(x) E(x) + \log Z_T,$$

which follows from the KL divergence with $E(x)$ replaced by $E(x)/T$. At high temperature T the distribution is smooth and easy to fit: for $T \rightarrow \infty$ every labelling has the same probability and the entropy term alone decides, so the beliefs are $1/2$ everywhere and $F[Q, T]$ has a single minimum. At low T it is peaked at the minima of E , and $F[Q, T]$ has the local minima of the original problem.

Minimise $F[Q, T]$ at large T , start the next smaller T from that minimiser, and repeat. The binary update at temperature T is $q_i = \sigma(-(\psi_i + \sum_j q_j \psi_{ij})/T)$, the sigmoid with its argument divided by T . The hope is that the minimum tracked from high temperature ends at the global minimum at low temperature; this is the continuation method, and it often avoids poor local minima without any guarantee of doing so.

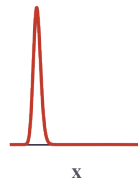
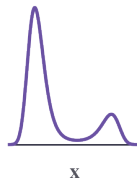
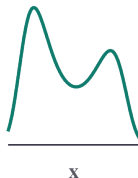
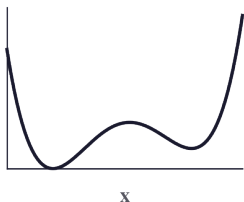
Deterministic annealing starts from a smoother distribution

energy E with two minima

$T=6$

$T=1.5$

$T=0.3$



$$P(x; T) = e^{-E(x)/T} / Z_T$$

smooth at high T , peaked at low T

Figure 35: An energy with two minima on one continuous variable, and the distribution $P(x; T)$ at three temperatures: smooth at high T , peaked at low T .

Summary

- ▶ A deep network gives unary evidence for the label at each pixel.
- ▶ A Markov random field adds pairwise potentials on the edges: the energy is evidence plus context.
- ▶ A conditional random field lets both terms depend on the image; a dense CRF weights every pair.
- ▶ The energy defines a Gibbs distribution, and the lowest-energy labelling is the most probable one.
- ▶ Enumeration is exponential in the number of pixels; exact inference on a loopy grid is intractable in general.
- ▶ Mean field theory approximates $P(x)$ by a factorised $Q(x)$ and minimises the free energy $F[Q]$.
- ▶ For binary labels the update is a sigmoid of evidence and neighbours' beliefs; for M labels it is a softmax.
- ▶ The concave-convex procedure explains the convergence, and annealing can help avoid poor minima.

References

- [1] P. Krähenbühl and V. Koltun. Efficient inference in fully connected CRFs with Gaussian edge potentials. *Advances in Neural Information Processing Systems*, 2011.
- [2] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy and A. L. Yuille. Semantic image segmentation with deep convolutional nets and fully connected CRFs. *International Conference on Learning Representations*, 2015.
- [3] J. Long, E. Shelhamer and T. Darrell. Fully convolutional networks for semantic segmentation. *IEEE Conference on Computer Vision and Pattern Recognition*, 2015.
- [4] A. L. Yuille and A. Rangarajan. The concave-convex procedure. *Neural Computation*, 15(4):915–936, 2003.